

Globs & wildcards

When you work with many Verilog files or log files, you do not want to type every name

Star, question mark, brackets

- Star matches any sequence of characters
- Question mark matches exactly one character, handy for data underscore question-mark log
- Brackets match one character from a set or a range, like a, b, or c in square brackets
- The shell expands these patterns first; the command only sees the resulting names

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Glob / wildcard lab

Expand *, ?, and [...] against a fixed directory — see what the shell passes to the command before it runs.

Starter example: Pattern *.txt expands to all .txt names. Try data_?.log (two files) vs data_*.log (three), and notice .gitignore is *not* matched by *.

Load starter example

Challenges 0 / 22 cleared

*.txt count: Expand *.txt. How many matches? (number)

Answer

Show hintCheckNextIdle

*.txt count

Expand *.txt

report_*.txt

data_?.log

data_*.log

Quiz: ? vs *

[abc].txt

[a-c].txt

*.sv

*.v

* skips dotfiles

.* for hidden

No match → literal

Quiz: when expand

Quiz: *

? .txt

Exact Makefile

.[sv] style

Show data_?.log

[!0-9] demo

Count non-hidden

Quiz: safe rm

Pattern → expansion

Glob pattern

Expand

Directory listing

highlight = match

a.txt

b.txt

c.txt

report_2023.txt

report_2024.txt

data_1.log

data_2.log

data_10.log

design.v

uart_tx.sv

th_top.sv

Globs and wildcards lab

learn_unix — wildcards globs

Real shell practice

```
wsl - module04-wildcards-globs/examples/wildcards

# Track A - real Unix
# real shell session (Track A)

$ ls
README.md
a.txt
b.txt
c.txt
data_1.log
data_2.log
design.v
report_2023.txt
report_2024.txt

$ ls *.txt
a.txt
b.txt
c.txt
report_2023.txt
report_2024.txt

$ ls report_*.txt
report_2023.txt
report_2024.txt

$ ls data_?.log
data_1.log
data_2.log

$ ls [abc].txt
a.txt
b.txt
c.txt
```

Real shell — globs and wildcards

Real shell practice — try these

```
# ls — see every name in this folder first  
ls
```

```
# ls *.txt — star matches any sequence; list all .txt files  
ls *.txt
```

```
# ls report_*.txt — star after a prefix; only report_... text files  
ls report_*.txt
```

```
# ls data_?.log — question mark matches exactly one character  
ls data_?.log
```

```
# ls [abc].txt — brackets: one character from the set a, b, or c  
ls [abc].txt
```

Pitfalls to watch

- The shell expands the pattern before the command runs
- If nothing matches, bash may leave the pattern literal
- Always list a destructive pattern first
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice star, question mark
- When you are ready, take the short quiz, then continue to file types and links